

Submission-Readiness Audit

Read against the repository as it stands on 8 September 2026: 184 measured runs, both baselines reproduced, the method implemented — and two defects that make every headline comparison provisional.

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Evidence	results/runs.csv (184 rows) + 12 run summaries	Date	8 September 2026

VERDICT

Not ready for submission. Two blockers, both fixable in one week of quota.

The engineering is in good shape. The toolchain works, both baselines run end to end on free compute, Mip-Splatting reproduces to within 0.26 dB at every scale, the seed spread is measured at $\sigma = 0.039$ dB, and the B1 filter is implemented correctly — including the non-commutativity of an anisotropic filter, which is handled through `cov3D_precomp` rather than the scale vector.

But the science is not yet established. The 3DGS arm is not 3DGS: it retains Mip-Splatting's 2D opacity compensation, so the measured baseline gap is 4.05 dB where the paper reports 10.98 dB. Gate G2 failed and the ladder continued anyway. And B1 has only ever been run on a full 179° orbit, where Proposition 2 *proves* it equals Mip-Splatting — which is exactly what the diagnostics report (`frac_above_floor = 0.0`, `mean_anisotropy = 1.0`). The decisive experiment of the thesis has not been run.

01 What has actually been done			
RUNG	STATUS	EVIDENCE IN THE REPOSITORY	ASSESSMENT
R0 smoke	DONE	r0_smoke_v2...v5 ; v5 = 01_t4x2_all_checks_pass.log	Four iterations including two p100_abort logs — the compute-capability guard fired exactly as specified. Pre-flight passed on v5.
R1 STMT	RUN, INVALID	64 rows, 8 scenes × 2 arms × 4 scales	Mip-Splatting arm excellent; 3DGS arm wrong (§02).
R2 MTMT	RUN, INVALID	64 rows, load_allres=True	Same defect. Mip-Splatting side within 0.54 dB.
R3 real	PARTIAL	8 rows — Mip-NeRF 360 indoor ×4 only	Tanks & Temples and Deep Blending not run. 3DGS/Mip-Splatting at parity (31.31 / 31.29), as expected indoors.
R5 seeds	DONE	32 rows, 2 scenes × 3 seeds × 2 arms	$\sigma = 0.039$ dB pooled, 0.139 dB worst case. Genuinely useful — parity is now testable at ± 0.12 dB (3σ).
R4 outdoor	NOT RUN	—	Optional by design. Correctly deferred.

I-1 floor	INSTRUMENTED	floor_diag/summary.json	Reports <code>frac_above_floor</code> and <code>mean_anisotropy</code> per run. Working — but only exercised where the answer is foreordained (§03).
I-3 spread	DONE	R5	See above.
I-2 $\wedge$ profile	PARTIAL	instruments/summary.json , tools/instruments.py	Instrument exists; λ_3/λ_1 never reported across capture protocols, because only <code>full</code> was ever built.
B1 method	NOT TESTED	scene/fisher_filter.py , t2_b1/summary.json	Implemented and trained (lego, chair, 30 K). But in the one regime where it provably cannot differ from the baseline.
B2 SH	NO RESULTS	kaggle/b2_eval.py , tools/sh_identifiability.py	Code written 8 Sep; no output directory, no Table 4.
Stress suite	NOT BUILT	tools/camera_protocols.py implements <code>full</code> , <code>arc</code> , <code>cone</code> , <code>grazing</code>	Only <code>full</code> was ever instantiated (<code>span_deg</code> $\approx$ <code>179°</code> , 100 cameras). <code>mixed_focal</code> not implemented.

Credit where due: `gaussian_renderer/__init__.py` routes the anisotropic filter through `cov3D_precomp` with the comment "*The filter is a matrix, so it cannot be folded into the scale vector the way a scalar can*", and adds a `filter_isotropic` fast path that takes Mip-Splatting's own code path so Proposition 2's parity is exact by construction. That is the correct architecture and it is the hardest part of the implementation.

02 Blocker 1 — the 3DGS arm is not 3DGS

2.1 · The measurement

TEST SCALE	3DGS OURS	3DGS PUB	Δ	MIP-S OURS	MIP-S PUB	Δ	GAP OURS	GAP PUB
Full res	33.31	33.33	-0.02	33.14	33.36	-0.22	-0.17	0.03
1/2	31.63	26.95	+4.68	33.83	34.00	-0.17	2.21	7.05
1/4	27.80	21.38	+6.42	31.77	31.85	-0.08	3.97	10.47
1/8	24.54	17.69	+6.85	28.59	28.67	-0.08	4.05	10.98

Blender synthetic, single-scale train $\rightarrow$ multi-scale test, mean over 8 scenes, seed 0, 30 000 iterations. Published column: Mip-Splatting Table 2.

GATE G2 FAILED, AND THE LADDER CONTINUED

G2 required the $\frac{1}{8}$ -scale gap to exceed **9 dB**. Measured: **4.05 dB**. The operating prompt said "*stop and do not proceed to R2*". R2, R3, R5 and the B1 method run were all executed afterwards. None of them is wrong in itself, but **every comparison against "3DGS" in the repository is currently a comparison against a different method**.

2.2 · The diagnosis, confirmed in source

The signature localises the fault precisely. Full resolution matches to **0.02 dB**, so the data, the split, the per-scale demultiplexing, the metric and the optimiser are all correct. Only the *degradation with scale* is wrong, and it is wrong in the direction of too much anti-aliasing.

gaussian_renderer/__init__.py passes kernel_size into GaussianRasterizationSettings — this is the **Mip-Splatting rasteriser**, which applies the 2D Mip filter *together with its opacity compensation* $p = \sqrt{(\det \Sigma' / \det(\Sigma' + kI))}$. Vanilla 3DGS adds a fixed 0.3f to the 2D covariance diagonal with **no compensation and no kernel_size parameter at all**.

arm B as run = 3DGS + the 2D Mip filter $\neq$ 3DGS

Setting kernel_size = 0.3 reproduces 3DGS's dilation *magnitude* while retaining Mip-Splatting's *anti-aliasing compensation*. That single term is exactly the configuration the Reproduction Report flagged under gsplat rasterize_mode="antialiased" — and it alone recovers most of the multi-scale robustness. Hence: identical at 1×, dramatically better at reduced scales.

2.3 • The fix

1. Add disable_2d_mip_compensation through GaussianRasterizationSettings to the CUDA rasteriser, returning the un-compensated fixed dilation when set — **or**, cleaner, build the stock graphdeco-inria/diff-gaussian-rasterization alongside and route arm B to it.
2. Re-run **R1** and **R2** for arm B only (arm A is unaffected and need not be repeated).
3. **Acceptance:** 3DGS full-res stays ≈ 33.3 and $\frac{1}{8}$ -scale falls to ≈ 17.7 . Then G2's 9 dB gap appears. Until both hold, do not proceed.

Cost estimate: 8 scenes $\times$ 2 protocols $\times$ ~45 min on 2×T4 $\approx$ 6–8 GPU-hours, well inside one week of quota. Arm A's 64 rows are already valid and are kept.

WHAT SURVIVES THIS BLOCKER INTACT

The **Mip-Splatting reproduction is excellent and is not affected**: -0.22, -0.17, -0.08, -0.08 dB against published on STMT, and -0.26 to +0.54 on MTMT. That is **L1 at three of four scales** and a marginal miss (0.22 against a 0.20 tolerance) at full resolution. The harness, the per-scale splitter, the VGG-LPIPS pinning and the multi-scale loader are all vindicated by it.

03 Blocker 2 — B1 has only been run where it cannot differ

Every B1 run in the repository reports the same two diagnostics:

RUN	SCENES	PROTOCOL	SPAN	FRAC_ABOVE_FLOOR	MEAN_ANISOTROPY
floor_diag	lego	full	178.98°	0.0	1.0
t2_b1	lego	full	178.98°	0.0	1.0
t2_b1	chair	full	178.66°	0.0	1.0

READ WHAT THESE NUMBERS SAY

frac_above_floor = 0.0 means the estimation term s^2/λ_i never exceeds the Nyquist floor f_k^2 , on any primitive. By **Proposition 2** that makes $\Sigma_{\text{filt}} = f_k^2 I$ identically — so **"B1" in these runs is Mip-Splatting under another name**, and mean_anisotropy = 1.0 confirms it. The t2_b1 numbers are not a method result.

3.1 · This is not a refutation — it is the predicted outcome, in the wrong regime

All three runs used `protocol: full`, a 179° orbit of 100 cameras. That is the *well-conditioned* case, where §4.6 of the thesis predicts $\lambda_3/\lambda_1 \approx 1$ and §4.7 *proves* reduction to the baseline. Obtaining exactly that is a **successful validation of Proposition 2**, not evidence against the method. It should be reported as such — it is the strongest possible confirmation that the implementation is correct.

What it is not is a test of the claim. Gate G3 asks whether the estimation term has room *anywhere*. Answering that requires the low-parallax and one-sided protocols, where theory predicts $\lambda_3/\lambda_1 \approx 0.003$ and 0.03 and depth-to-lateral ratios of 19× and 5.8×.

PROTOCOL	STATUS	PREDICTED λ_3/λ_1	WHAT RUNNING IT DECIDES
full (control)	RUN	≈ 1	Done, and it passed. Proposition 2 validated empirically.
cone low-parallax	NOT RUN	≈ 0.003	The decisive experiment of the thesis. Largest predicted gain.
arc one-sided	NOT RUN	≈ 0.03	Intermediate gain — confirms the <i>ordering</i> , which is what makes the regime structure a result rather than a single point.
grazing	NOT RUN	—	Anisotropy aligned with the surface normal.
mixed_focal	NOT IMPLEMENTED	—	Tests the floor's mixed-camera defect (§3.3 of the thesis). Needs writing.

Without the cone and arc rows there is no thesis result — only a correctness proof and a reproduction. This is the single highest-priority remaining task, and at ~1.8 GPU-hours per 2-scene protocol it costs roughly 8–10 GPU-hours for a 4-protocol × 2-scene × 2-method grid.

04 What is genuinely established

Stated separately so it is not lost among the defects. These results are sound and can go into the thesis now.

RESULT	LEVEL	EVIDENCE
Mip-Splatting reproduces	L1 / L2	STMT $\Delta = -0.22, -0.17, -0.08, -0.08$ dB; MTMT $\Delta = -0.26, -0.06, +0.32, +0.54$ dB. Independent confirmation of the published tables on 16 GB free compute.
Seed spread	MEASURED	$\sigma = 0.039$ dB pooled, 0.139 dB worst case, over 3 seeds × 2 scenes × 2 methods. Parity is now a testable claim at ± 0.12 dB.
Proposition 2 holds empirically	VALIDATED	<code>frac_above_floor = 0.0</code> and <code>mean_anisotropy = 1.0</code> on the full orbit, with <code>propositions_pass: true</code> in every summary. The reduction is exact in practice, not only on paper.
Anisotropic filter is implementable without CUDA changes	DEMONSTRATED	Routed through <code>cov3D_precomp</code> ; trains to 30 K, 271–343 K primitives, 13.7–14.1 GB peak VRAM — inside the 16 GB envelope.
Free-tier feasibility	DEMONSTRATED	Whole programme executed on 2×T4 with checkpoint/resume; P100 guard fired correctly twice; ~1.8 GPU-h per 2-scene method run.

PART II · CORRECTIONS TO THE THESIS

05 Mathematical gaps found on re-reading

Independent of the experiments. Each is a place a careful examiner would push, with the correction to apply.

M-1 · The visibility weight w_{nk} is never defined

The thesis writes $\Lambda_k = \sum_n w_{nk}(\dots)$ and calls w_{nk} "the blending contribution the rasteriser already accumulates", but never defines it or justifies it. A weighted sum of Fisher informations is not itself a Fisher information unless the weights are counts. **Correction:** define $w_{nk} = T_{nk}\alpha_k \in [0,1]$, the accumulated transmittance- weighted contribution of primitive k in view n , and state plainly that this is a *heuristic down-weighting of occluded views*, not a likelihood-derived quantity — so Λ_k is an observability matrix that coincides with the Fisher information when all weights are 1. Honest, and it removes the only soft step in the derivation.

M-2 · s and σ_{pix} are over-determined

§4.4 says " s is a confidence scaling, default 1"; Proposition 1 requires $s \cdot \sigma_{\text{pix}} = \sqrt{0.2}$. Both cannot hold with σ_{pix} free.

Correction: fix $\sigma_{\text{pix}} = 0.447$ px by inheritance from Mip-Splatting's calibration and let s be the dimensionless confidence multiplier with default 1, so that $s = 1$ is exactly the baseline's calibration. One sentence, and the ablation over s becomes interpretable as "how many baseline-confidence radii".

M-3 · The Cramér–Rao framing overclaims

The CRB bounds unbiased estimators of a point parameter; the optimiser is neither unbiased nor estimating a point.

Correction: present Λ_k as the *Gauss–Newton approximation to the Hessian of the photometric-consistency cost with respect to primitive position, under a point-feature approximation*, and keep the CRB as the motivating interpretation. Strictly more defensible, and costs nothing.

M-4 · A tempting simplification that is wrong — worth pre-empting

For a camera aimed at the primitive, I_n reduces exactly to $(f/z)^2/\sigma_{\text{pix}}^2 \cdot (\mathbf{I} - \mathbf{d}\mathbf{d}^\top)$, a rank-1 update needing no Jacobian.

It is tempting to implement that. **It is only valid on-axis.** Verified numerically in general position: for an off-axis primitive the two non-zero eigenvalues are *not* equal — λ_2/λ_1 measured at 0.512, 0.682, 0.956, 0.997 across four random poses. So perspective contributes anisotropy *on its own*, before any parallax argument.

THIS STRENGTHENS THE THESIS

There are **two** independent sources of anisotropy: parallax (the $1/\sin(\theta/2)$ result) and off-axis perspective (up to $\approx 1.4\times$ from a single view). A scalar filter discards both. Add this as a short subsection with the four measured ratios — it is a new argument, it costs no GPU time, and it pre-empts the reviewer who proposes the rank-1 shortcut.

M-5 · Appendix B.2 can be exact rather than descriptive

The current proof gestures at the projection. Since each view aimed at the primitive contributes $(f/z)^2/\sigma^2 \cdot (\mathbf{I} - \mathbf{d}_n\mathbf{d}_n^\top)$, two symmetric views give $\Lambda \propto 2\mathbf{I} - \mathbf{d}_1\mathbf{d}_1^\top - \mathbf{d}_2\mathbf{d}_2^\top$, diagonal in the natural frame with entries

$$\lambda_{\text{out-of-plane}} = 2, \quad \lambda_{\text{in-plane}} = 2\cos^2(\theta/2), \quad \lambda_{\text{depth}} = 2\sin^2(\theta/2)$$

giving $\lambda_3/\lambda_1 = \sin^2(\theta/2)$ in two lines. **The middle eigenvalue $2\cos^2(\theta/2)$ is a new result** and should be stated: the in-plane lateral direction is also degraded, so the filter is triaxial rather than merely "long in depth". Verified against full pinhole Jacobians at $\theta = 2^\circ, 6^\circ, 15^\circ, 45^\circ, 90^\circ, 120^\circ$ with maximum deviation 6×10^{-11} .

M-6 • The opacity compensation deserves a one-line lemma

Reusing Mip-Splatting's α -compensation for an *anisotropic* filter is asserted, not justified. It is in fact exact: total mass $\alpha \cdot (2\pi)^{3/2} |\Sigma|^{1/2}$ is preserved under $\Sigma \rightarrow \Sigma + \Sigma_{\text{filt}}$ for any positive-definite Σ_{filt} . Verified to machine precision on random anisotropic pairs. State it — it is free rigour.

M-7 • The SH threshold τ has no scale

A_k grows with the number of views, so a fixed τ means different things for different primitives. **Correction:** normalise, $\bar{A}_k = A_k / \Sigma_n w_{nk}$, and threshold $\lambda_{\min}(\bar{A}_k^{(\ell)}) > \tau$.

M-8 • Implementation consequence to state in the thesis

An anisotropic Σ_{filt} does not commute with Σ_k , so Mip-Splatting's per-axis $\sqrt{(s^2+f^2)}$ update is *not* available — the full 3×3 sum must be passed to the rasteriser. The repository already does this correctly via `cov3D_precomp`; the thesis should say so, because it is the one place where "a matrix, not a scalar" has a real engineering cost. Also state that Δ is computed under `no_grad` and held constant between recomputations, exactly as `filter_3D` is — which is what removes any question of differentiating an eigendecomposition.

PART III • THE PATH TO SUBMISSION

06 Remaining work, ordered

#	TASK	GPU-H	EXIT CRITERION
C1	Fix the 3DGS arm. Disable the 2D Mip opacity compensation, or build the stock Inria rasteriser for arm B.	0	A one-scene check: full-res ≈ 33.3 and $\frac{1}{6} \approx 17.7$.
C2	Re-run R1 and R2, arm B only. Arm A's rows are valid and are kept.	6–8	G2: $\frac{1}{8}$ -scale gap > 9 dB. Table 1 becomes real.
C3	Build the stress protocols — <code>cone</code> , <code>arc</code> , <code>grazing</code> exist in <code>camera_protocols.py</code> ; instantiate them and write <code>mixed_focal</code> .	0	Four protocols versioned as seeded JSON index lists, with measured <code>span_deg</code> per scene.
C4	Run I-2 across protocols. λ_3/λ_1 on trained baseline checkpoints — no retraining, no method code.	1–2	G4. The regime table becomes measured. This is the mid-term seminar figure.
C5	The decisive experiment. B1 vs Mip-Splatting on <code>cone</code> , <code>arc</code> , <code>grazing</code> , <code>mixed_focal</code> , 2+ scenes.	8–12	G3. <code>frac_above_floor</code> > 0 and $\Delta > 3\sigma = 0.12$ dB in the predicted ordering, or the claim narrows.

C6	Merge B1 rows into <code>results/runs.csv</code> ; regenerate all tables and figures from it.	0	Every thesis number traces to one CSV row. Currently B1 lives only in <code>t2_b1/summary.json</code> .
C7	B1 seed spread — 3 seeds on one stress protocol.	3–4	σ for the method, so the gain is reported as a multiple of it.
C8	B2 evaluation — run <code>b2_eval.py</code> , produce Table 4 against magnitude pruning at matched size.	4–6	Table 4 populated; the 76 % claim becomes a measurement.
C9	Matched fps measurement, three methods, same scene and GPU in one run.	<1	Current <code>render_fps</code> of 8–10 is not comparable to the published 180–300 and almost certainly includes I/O. The claim is <i>unchanged relative to baseline</i> , so measure the ratio.
C10	Complete R3 — Tanks & Temples and Deep Blending.	8–10	Reference values already in the thesis (DB 29.41, T&T 23.14) become comparisons.
C11	Apply M-1...M-8 to the thesis text.	0	Zero-GPU. Can be done today, in parallel with everything above.
C12	§08 literature due-diligence — has anyone already made the 3D filter anisotropic or per-direction? FisherRF and PUP 3D-GS in particular.	0	The only task that can invalidate the project. Still no evidence of it in the repository. Zero-GPU, and overdue.

Total $\approx$ 30–43 GPU-hours — one to one-and-a-half weeks of a 30 h quota. C1–C5 are the critical path and are worth roughly 15–22 of those hours; everything else can follow or run in parallel.

THE HONEST SEQUENCING

C1 and C2 first, because every table that mentions 3DGS is provisional until they land. C3–C5 next, because they are the only experiments that can turn a correctness proof into a finding. C12 in parallel from today — it costs nothing but time and it is the one result that could make the rest moot.

07 Submission-readiness matrix

CRITERION	STATUS	ASSESSMENT
Theory complete and internally consistent	YES	Two propositions, one closed form, all verified numerically. M-1...M-8 are refinements, not repairs.
Method implemented and correct	YES	Anisotropic path via <code>cov3D_precomp</code> ; isotropic fast path makes Proposition 2 exact by construction; <code>propositions_pass: true</code> throughout.
Baseline B reproduces	YES	Mip-Splatting within 0.26 dB at every scale, both protocols.
Baseline A reproduces	NO	Blocker 1. 3DGS arm retains the 2D Mip compensation; gap 4.05 dB against a required 9.
Method evaluated against baselines	NO	Blocker 2. Only the regime where Proposition 2 forces equality.
Central claim tested	NO	Stress suite not built. λ_3/λ_1 never measured away from a full orbit.
Error bars	YES	$\sigma = 0.039$ dB pooled. Better than the project needed.

Ablations	PARTIAL	The floor-vs-no-floor and isotropic-vs-anisotropic ablations follow directly from C5; the s and β sweeps do not exist.
B2 result	NO	Code only.
Literature due-diligence	NO	Not evidenced. The single largest un-retired risk in the project.
Reproducibility discipline	YES	184 rows with commits, flags, VRAM and level claimed; per-rung logs and summaries; P100 aborts preserved rather than deleted. Exemplary.

BOTTOM LINE

This is a project with **excellent infrastructure and an untested central claim**. Nothing discovered in this audit threatens the theory; both blockers are configuration and coverage problems, not conceptual ones, and both are fixable inside a week of free-tier quota. The most valuable hour available right now is not a GPU hour — it is the §08 literature check (C12), which has been outstanding since week 1 and is the only thing that could make the other forty hours unnecessary.

Scope of verification. All figures in §01–§04 were computed directly from `D:\BTP\repro\results\runs.csv` (184 rows) and the twelve run summaries under `results/kaggle_runs/`, staged 8 September 2026. The diagnosis in §02.2 was confirmed by reading `gaussian_renderer/__init__.py`. The claims in M-4, M-5 and M-6 were verified numerically in this session against full pinhole Jacobians and random anisotropic covariance pairs. Published comparison values are from Mip-Splatting (arXiv:2311.16493) Tables 1–2 and the 3DGS paper as recorded in Handbook §02.